

CSA0613 - Design and Analysis
of Algorithms

CO2-AT1 - Analytical Problem Solving

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Bubble Sort - Duplicate Values Analysis

Problem Interpretation :

Given the dataset :

$[32, 14, 32, 7, 19, 7]$

Sort the elements in ascending order using the Bubble sort algorithm and analyze its performance.

Why Bubble Sort works with Duplicate values?

- * Bubble sort compares adjacent elements and swaps them only if the left element is greater than the right element.

- * If two adjacent elements are equal, no swap is performed.

- * Therefore, duplicate values remain in their relative order, making Bubble Sort a stable sorting algorithm.

Initial Array :

$[32, 14, 32, 7, 19, 7]$

Pass 1 :

Compare 32 and 14 $\rightarrow$ Swap

$[14, 32, 32, 7, 19, 7]$

Compare 32 and 32 $\rightarrow$ No swap

$[14, 32, 32, 7, 19, 7]$

Compare 32 and 7 $\rightarrow$ Swap

[14, 32, 7, 32, 19, 7]

Compare 32 and 19 $\rightarrow$ Swap

[14, 32, 7, 19, 32, 7]

Compare 32 and 7 $\rightarrow$ Swap

After pass 1 : [14, 32, 7, 19, 7, 32]

comparisons = 5

Swaps = 4.

Pass 2 :

Compare 14 and 32 $\rightarrow$ No Swap

compare 32 and 7 $\rightarrow$ Swap

[14, 7, 32, 19, 7, 32]

Compare 32 and 19 $\rightarrow$ Swap

[14, 7, 19, 32, 7, 32]

Compare 32 and 7 $\rightarrow$ Swap

[14, 7, 19, 7, 32, 32]

After Pass 2 :

[14, 7, 19, 7, 32, 32]

Comparisons = 4

Swaps = 3

Pass 3 :

compare 14 and 7 $\rightarrow$ swap.

[7, 14, 19, 7, 32, 32]

compare 14 and 19 $\rightarrow$ No swap.

compare 19 and 7 $\rightarrow$ swap.

[7, 14, 7, 19, 32, 32]

After Pass 3 :

[7, 14, 7, 19, 32, 32]

comparisons = 3

swaps = 2.

Pass 4 :

compare 7 and 14 $\rightarrow$ No swap

compare 14 and 7 $\rightarrow$ swap

[7, 7, 14, 19, 32, 32]

After Pass 4 :

[7, 7, 14, 19, 32, 32]

comparisons = 2

swaps = 1.

Pass 5 :

Compare 7 and 7 $\rightarrow$ No swap

After Pass 5 :

[7, 7, 14, 19, 32, 32]

Comparisons = 1

Swaps = 0

Total comparisons : 15

Total Swaps : 10

Final Sorted Output :

[7, 7, 14, 19, 32, 32]

Time Complexity Analysis :

* Best case : $O(n)$

* Average case : $O(n^2)$

* Worst case : $O(n^2)$

Correctness Verification :

The elements are arranged in ascending order :

$7 \leq 7 \leq 14 \leq 19 \leq 32 \leq 32$

Hence, the sorted output is correct.

Behaviour on Duplicate Elements :

- * Bubble sort correctly handles duplicate values.
- * Equal elements (32, 32 and 7, 7) are not swapped with each other.
- * Their original order is preserved, showing that Bubble sort is a stable sorting algorithm.

Result :

The given dataset [32, 14, 32, 7, 19, 7] was successfully sorted using Bubble sort. The final sorted array is [7, 7, 14, 19, 32, 32] with 15 comparisons and 10 swaps. The algorithm correctly handles duplicate values while preserving their relative order.